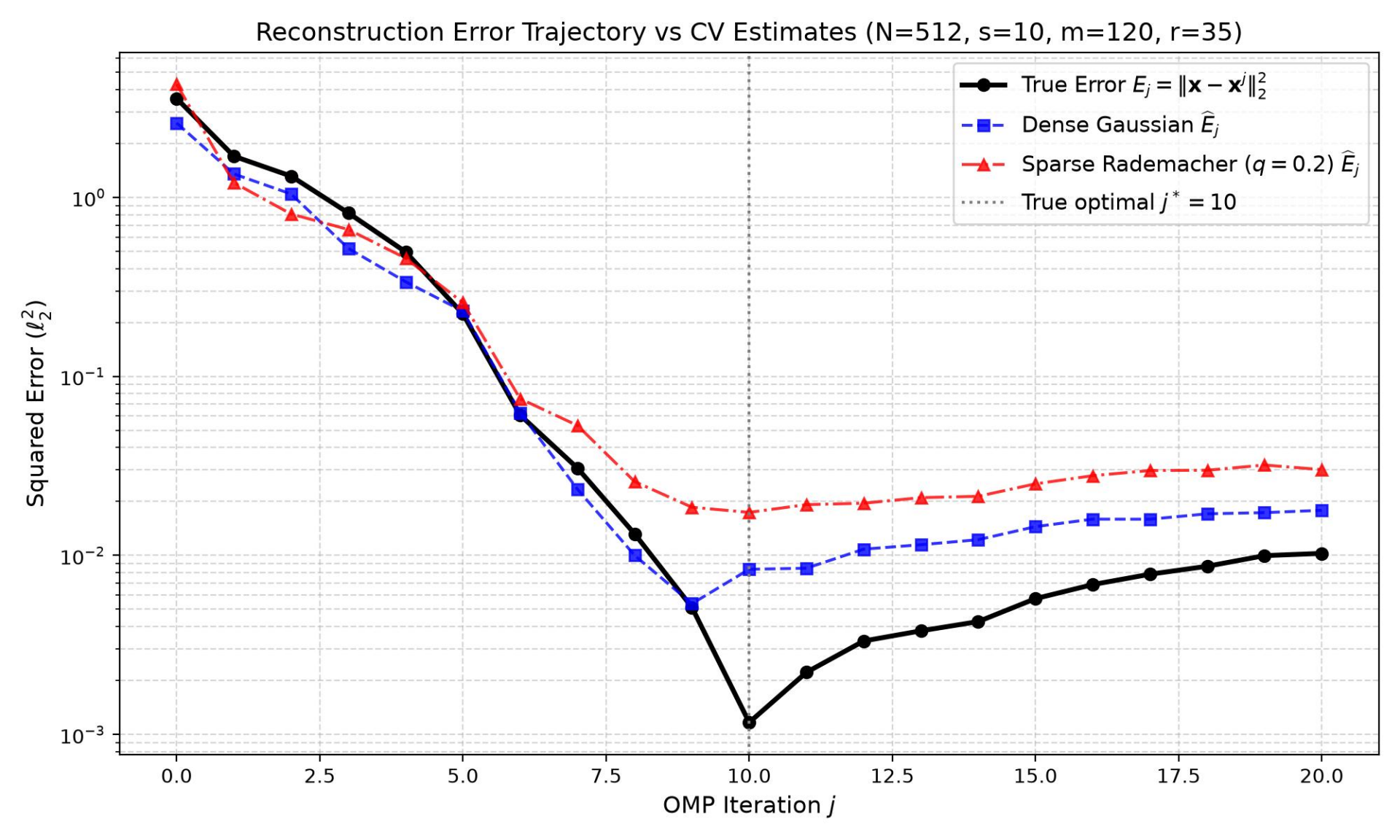

Reconstruction Error Trajectory vs CV Estimates (N=512, s=10, m=120, r=35)
True Error $E_j = \|\mathbf{x} - \mathbf{x}^j\|_2^2$
Dense Gaussian $\widehat{E}_j$
Sparse Rademacher ($q = 0.2$) $\widehat{E}_j$
True optimal $j^* = 10$
Squared Error (ℓ_2^2)
10^0
10^{-1}
10^{-2}
10^{-3}
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
OMP Iteration j